

Senior Enterprise Architect

Distinguished Architect, Project Manager & Technical expert, Result orientated team leader with extensive skills in the area of Application & Integration, Cloud Computing/Orchestration, Networking, Business technology alignment and Program management. With 20+ years of proven hands on delivery, strategic guidance, technical design and leadership in some of Australia's leading IT projects.

I have built and designed redundant and cost effective ERP, HR, POS systems & various other business critical delivery systems to address and support complex business requirements for small-to-large Enterprises across Australia and the World. I have been involved from full life cycle business analysis, market engagement, to Storage/Network/Application delivery giving me a broad understanding of all underlying requirements; this experience greatly assists in a coherent solution delivery.

I have years of experience starting as a technical expert to a seasoned Enterprise Architect with over a decade of Architecture design & Program management. My focus is delivering business driven change through effective IT service based solutions.

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|-------------------------|-------------------------|---------------------------|
| ◆ Cloud Architecture | ◆ Network Architect | ◆ Infrastructure Design |
| ◆ IT Strategy | ◆ ERP Architecture | ◆ Operations Optimisation |
| ◆ Security Architecture | ◆ Program Management | ◆ Change Management |
| ◆ IT Architecture | ◆ Software (SOA) Design | ◆ Business Architecture |
| ◆ SCADA Architecture | ◆ Payroll Systems | ◆ Large Scale Deployments |
| ◆ Database Architecture | ◆ Strategic Planning | ◆ Systems Integration |

Professional Experience

EXTREME-E

LONDON, UK

ENTERPRISE/CLOUD ARCHITECT – PROGRAM MANAGER

NOVEMBER 16TH 2019 – FEBRUARY 12TH 2021

The initial contract was to do a complete analysis of the systems and architecture that would be required to run the custom race applications and media systems. After completing a current state my contract was extended and I was tasked with developing the business case and market approach for the future state architecture, roadmap and environment to meet the launch date of Extreme-E.

This was a significant task engaging with up to a dozen key business stakeholders to ensure the system was both functional, manageable and fit for purpose; another important factor was the system had to be easily scaled up & down with the mandate that the system be as eco-friendly by using as minimal energy as possible. I did all the tender requirements and successfully went to market resulting in two substantial partners being used to develop an industry leading platform.

- Current State Architecture, Future State Architecture;
- ERP – SAP/GIS Solution. Full Geospatial integrated SAP Solution
- Development system Architecture (DevOP);
- Multi-Cloud Platform with centralized data and deployment base;
- Secure real time content distribution;
- Remote Cloud based workflows;
- Multi Database systems Inc. MSSQL & Proprietary unstructured AI R2DP System;
- Office365 Integration;
- Tender preparation, Go to Market and Evaluation;
- RESTFUL Services – Considerations SPA - Angular, React & Backbone;
- Agile Project Management;

AJILON & ADECCO

BRISBANE, QLD, AUSTRALIA

PROGRAM MANAGER/SOLUTION ARCHITECT - CONSULTANT

QLD Health

Engaged by QLD Health to project manage and design a secure cloud based video conferencing solution for Patient – Doctor Consultations.

The solution was designed for maximum security due to the sensitivity of the traffic traversing and availability as it is required due to strict timing.

I was responsible for the entire design with the key some of the components and following requirements:

- Networking: Work across all internet connections and enable end-to-end encryption.
- Infrastructure: Must leverage cloud where possible and without jeopardizing security.
- Security: Must meet or exceed QLD Government Security Framework.
- Integration: Must integrate with current Doctor Video/Telephony solutions.

QLD Department of Science, Information Technology and Innovation

(Limited Information Due to NDA & Clearance)

Responsible for the design and documentation of a QLD Government One-Stop Shop Initiative to make public services simpler, clearer and faster.

The solution has multiple integration points into various different Government Departments such as Department of Transport and Main Roads, Department of Communities, Child Safety and Disability, Department of Treasury and Trade.

The solution allows for a QLD Customer to be authenticated once with a complex SSO solution and then able to securely access forms that change information across a selection of departments at once.

The integration solution uses various technologies all cloud based:

- Mulesoft ESB
- Kiteworks
- Intelliforms
- F5, Reverse Proxy's, Cisco, Azure LB

DX SOLUTIONS & DIRECT CONSULTING

SYDNEY, NSW, AUSTRALIA

Technical Architect/Project Manager

T-Bone & Toyota Australia

Engaged by T-BONE to lift and shift the Toyota AUTOCONSOLE 3.0 onto Australia's Microsoft Azure platform creating a highly available modulated foundation that could handle millions of hits a month and easily scale on the fly. It was also to be enabled for continuous integration and development, Hooks into Jenkins and Bitbucket with auto deployment to Azure.

Built complex puppet scripts to automate the launch and scalability of the machines and load balancers required to deliver the solution. Utilised the latest features of Azure that are only available via Azure Powershell this included full Network Isolation and site to site VPN's for secure access and development. The site is spread across the two Sydney availability zones in Microsoft SQL and IIS availability groups creating an extremely redundant and highly available foundation.

The solution was delivered in a tight time frame on budget and to design, since launch it has not failed and requires little ongoing management.

Tools: Jira, Git, BitBucket, Bamboo, Puppet

Privatecall.me

Engaged to build a highly available private end to end encrypted telephony system on Microsoft Azure and AWS.

Designed and built the backend system on Chef Scripts using Linux SIP servers and Microsoft IIS front end servers with transparent encrypted data both in transit (SQL) and block storage.

I designed the DevOps CI/CD Architecture using GitHub – BitBucket and Jenkins for CD into EC2, I would also hook into S3 storage for various backups of sets.

TRIFORCE SERVICES PTY LTD

SYDNEY, NSW, AUSTRALIA

3 Month contact extended.

Chief Cloud Architect & Senior Engineer

May 2016 – February 2018

Triforce is dedicated to architecting and delivering sophisticated infrastructure solutions and to resolving complex technical issues as well as helping to align your IT to your business objectives.

Triforce is also one of Australia's leading suppliers of computing hardware, software and networking IT technology, in particular, one of the top 10 HP Premier Partners in Australia.

- NSW Government First Secure Platform as a Service; Responsible for the complete system and delivery architecture of the NSW Government Secure PaaS environment within Metronode government DC HALL Silverwater. Developed complex architectural designs around compute, network and orchestration to ensure security and isolation was of the highest standard. Once designs approved from Government, Sec-logic and Triforce. I then built the entire platform and delivered within time and budget. Due to the complexity, maturity and level of security required for the environment I was the only resource to design and build. I also designed the distributed networks and virtual networks / storage delivery for cross site redundancy and failover, though the redundant site is yet to be built I have documented and prepared for the entire solution from initiation. The solution is currently in use across the Government delivering applications from Oracle to 3sis and general web based delivery.
- Designed and built a complex Apache CloudStack platform utilising the latest technology's in network virtualization across multiple hypervisor mechanisms. I believe the first deployment of this scale within Australia on the latest Apache CloudStack for any private business utilising network virtualisation only and multiple hypervisors. I am a leading blog contributor and influence for bugs within the ACS community. I am also an active contributor/member of the Apache member's committee which leads discussion around new releases.
- Designed and built multiple high availability 99.999x PaaS environments from VMware, HyperV, Cloudstack and the latest HP Helion. Latest design was a cutting edge large scale PaaS solution for Papa New Guinea's Digi cell communications. The design was based on a multimillion dollar HP Helion solution. HP commissioned me via Triforce due to lack of internal capability to deliver the solution architecture. The design also included network components such as Juniper, F5 and Cisco.
- MVRC (Moonee Valley Race Club) Network upgrade. Major network upgrade utilising latest HP IRF technology (A5120) for L2 Redundancy without the hassle of STP. Design and deployed over 32+ switches across MVRC for a total site redundancy as required during the spring festivals including the legendary Cox Plate. Complete fibre backplane with Cisco ASA and Cyber-roam for firewalls, delivering and maintain a large scale wireless network utilising Ruckus. Total site wireless capability of 25,000 simultaneous users.

GOLD COAST CITY COUNCIL

GOLD COAST, QLD, AUSTRALIA

4 Month contract extended to 12 Months (ERP Delivery).

Solution Designer / Project Management

Nov 2014 – May 2016

City of Gold Coast is the local government area spanning the Gold Coast, Queensland, and surrounding areas. Based on resident population, it is the second largest local government area in Australia and its council maintains a staff of over 2,500

Solution designer for multiple projects including

- Enterprise Wireless Solution; Project Manager and lead solution designer for enterprise wide wireless solution. Wireless solution was designed with full high availability and redundant site disaster recovery, utilising world class Cisco Identity Service Engine and Policy servers; access points and wireless controllers to deliver a best practice solution to facilitate ERP mobility, corporate access and guest internet. Due to the Enterprise Wireless system being a dependency for the ERP SAP Phase II launch, it had an incredibly strict time frame that had to be met. I was able to deliver the project on time and budget assisting our vendor and in-house technical crew to achieve a great result for the business.
- Functional and non-functional design input for business case development and feasibility study on Laboratory Information System (LIMS) project. Commercial off the shelf (COTS) solution design and cloud connectivity modelling. Design considerations through all underpinning infrastructure layers with a significant view for security and exposure to Council.
- SQL as a service (SQLaaS) delivery model internal to Gold Coast City Council (GCCC), develop solution for infrastructure from storage to application layer. Build an ITIL template and support model for SLA management of the end to end service, considering the solution will underpin multiple business services. The first as a service modelling within GCCC for reduction of Microsoft core licensing, consolidation and simplification of management for over 500 disparate SQL databases.
- Microsoft Project Server 2010 project managed, solution designed and documented a large scale Microsoft Project 2010 server platform for utilisation across GCCC. The project server had 3x environments dev, test and prod each with x integration connectivity to SAP. I was lead architect for design, specifications, functional and non-functional designs and SAP integration. The project was successfully delivered on time to meet the Windows 7 rollout.
- ERP Implementation recovery; Engaged to ensure business requirements were met effectively across whole of Council. Managed ESRI's development and deployment whilst minimising customisation assisting in lowering IT across in the future. Was brought in to assist with project recovery after significant delivery delays.

REZILIENT IT

BRISBANE, QLD, AUSTRALIA

Technical Architect

Jan 2012 – Dec 2013

Rezilient IT develops robust solutions for Small; Medium Enterprises based on multiple IaaS, SaaS & PaaS cloud providers. Rolling out in house developed solutions we are able to build secure, cost effective highly available solutions on multiple providers ensuring reliable connectivity and latency across Australia.

A couple of examples of what I delivered:

- 1) Australian Embassy in Jakarta where we secured permission to utilise Amazon's Web Services from the Singapore region to deliver a load balanced virtual desktop solution with connectivity back to Australia. This is now in production as of August and will remain the source of truth and connectivity for the entire project, which is the largest Australian Embassy to be built. The design was built on custom architecture with underlying services from AWS. The architecture was sold to RightScale partners of AWS for future development of the service. Since I note that the IPSEC VPN functionality is now very

similar to our custom built linux servers.

- 2) Thinc Projects ICT 2012 strategy consisting of an entire ICT platform shift from build own operate to an outsourced approach leveraging multiple cloud providers across a large geographically disparate landscape. Consolidation of over 40 servers into a managed console offering scalability, flexibility and cost effective delivery from anywhere in the world. Included a large 300+ user upgrade from 2003 exchange through multiple steps to finally a complete organisation upgrade to 2013 exchange. Microsoft Dynamics AX shift to the cloud. IT Strategy available as reference.
- 3) Large internal multi-site multi-tenant environment. Designed and implemented a large PaaS/IaaS platform using Microsoft Hyper-V, VMWare for the virtualization layer including SMB3 continuous file access cross site cluster. The underlying physical network was built on Cisco switches and Cisco Routers / ASA devices, we also used Juniper for remote access, BigIP F5 devices for load balancing across legacy applications. I developed a custom Linux SNORT server using several open source solutions to deliver a powerful clustered IPS/IDS. Cisco Ironport cluster handled the large amounts of inbound & outbound email filtering.

QUEENSLAND URBAN UTILITIES LTD

BRISBANE, QLD, AUSTRALIA

A water and wastewater business established as a result of the Queensland State Government's changes to the way water is managed in South-East Queensland. Queensland Urban Utilities (QUU) is among the largest water and wastewater retail-distributors in Australia, supplying water and wastewater services to the cities and townships of Ipswich, Lockyer Valley, Scenic Rim, Somerset and Brisbane regions.

Principal Architect & IT Program Manager

May 2010 – July 2012

Initial 3 month contract (plus additional extensions). Continued advising into 2013.

Principle Architect and member of IC&T Leadership team developing foundation principles and frameworks to deliver structured approaches to Cloud Computing, Security, Network and VoIP architecture solutions for whole of organisation. This included significant designs for ERP, Identity management, Active Directory, Single Sign-on and Active Directory forest trust with Brisbane City Council.

Created an Enterprise Architecture for the organisation based TOGAF, creating foundation papers for the IC&T Strategy, procuring infrastructure solutions from IT Vendors, evaluating current and future program roadmaps, leading onsite IT development projects in a Project Architecture role and defining solutions for the business.

All Architectural designs where a 3 phase approach;

- Technical Separation Architecture;
- Transitional State Architecture; and
- End State Architecture,

This was due to a mandate to technically separate from Brisbane City Council (BCC) our IC&T vendor by December 2012.

Principal Network Architect on ERP Ellipse/ESRI Lift – Redesign/Adjust – Shift, LAN/WAN/MAN implementation for Corporate, SCADA, CCTV Networks, utilising Private Radio, 3G/4G, Fibre, DSL and Satellite. The design was Greenfield and had to support separation, transition and end state with continuity of service and business as usual operations on critical Government infrastructure. Five year projected planning with a budget of over \$15 Million dollars. I designed and project managed the entire solution from Tender/Procurement to Fibre builds to cut over including two diverse path L2 fibres from QLD to Sydney Fujitsu DC.

Implemented an 'end to end' SLA for all sites on both Corporate and SCADA WAN's, this included 99.99% uptimes across fixed line and wireless sites with on the job commitments of 15 minutes and uptime of a max 4 hour period.

Developed standards and guidelines for infrastructure platforms to cater for over 1,000 sites and 1,200 staff. These included activities such as presenting Solution Architecture designs to Executive Management to obtain project funding for the development and implementation of solutions.

Reviewed and evaluated business requirements to align with strategic architectural direction.

Lead technical discussions with service providers on behalf of QUU for example; ERP, GIS, WAN/LAN topologies, Corporate VoIP design, HR & Payroll (HRIS), SCADA/SCADAaaS HMI redesigns for end-to-end solutions and Application integrations.

Developed ICT Security and Network Architecture guidelines and principles to be provided to 3rd party vendors.

IC&T Program of Works redevelopment and Business Case Development.

Solutions for Man Down Alarms (MDA), Trunk Radio.

Complete Australian Market scan with Ernst & Young to assess and report on the feasibility of Cloud Computer for Government Critical Infrastructure.

PROVISIONAL ELECTRICAL AUTHORITY

Bangkok, Thailand

The Provincial Electricity Authority (PEA) is a government enterprise and are primarily concerned with the generation, distribution, sales and provision of electric energy services to the business and industrial sectors as well as to the general public in 73 provincial areas, with the exception of Bangkok, Nonthaburi and Samut Prakran provinces, covering approximately 510,000 km², accounting for 99% of the country's total area. PEA is headquartered in Bangkok, Thailand with annual revenues over AUD \$8.4 billion, with staff of 26,000.

Cloud Computing Architect / Program Manager

Jan 2009 – Feb 2011

Initial 6 month contract (plus additional extensions).

Lead Cloud Compute Architect.

ERP Business case, tender documentation and evaluation. Managed service engagement and program delivery client side.

Designed cloud based solutions to migrate 200+ Servers which incorporated client and server applications across SaaS, PaaS and IaaS platforms.

Re-align large scale (1000+ devices) WAN architecture and Security architecture to support cloud based integration and service offerings.

Cloud based routing, disaster recovery, IDS, IPS and load balanced trafficking.

Implemented technical end-to-end solutions in aggressive timeframes to achieve cost savings of over USD \$4 million.

Presented Solution Options inclusive of recommendations based on Business Strategy and IT Architectural alignment to the CEO and Board members.

Lead technical discussions in cloud solutions to migrate legacy systems and in-house developed applications into a Highly Available and Redundant platform.

Created windows servers with our cloud providers to configure full automation of deployments, from a new server launch to a fully function server with the application including (auto) scaling and load balancing.

Automation of server deployments/instances using PowerShell to configure a baseline server through to a functional application environment.

Managed 200+ servers through the Rightscale Management using the Amazon AWS API.

TOURAUST CORPORATION

Sydney, NSW, Australia

Touraust Corporation Pty Limited is a wholly Australian owned company that operates the following businesses

- International College of Management Sydney (ICMS) In association with its strategic partner Macquarie University, ICMS offers diplomas as well as associate degrees, bachelor's degrees and master's degrees in areas the of hospitality management, event management, international tourism, retail services management, sport management and property services management.
- Constellation Hotels is one of the most dynamic and rapidly growing hotel networks across Australia and New Zealand with close to 70 locations, Owner of five brands; Chifley, Grand Chifley, Country Comfort, Australis and the SunDowner.

Chief Information Officer (CIO)

Jan 2007 – Jan 2009

Fulltime Employee

Budget \$10 million.

Responsible for all aspects of Information Technology for the College and Hotel Group, including IT Strategy, Staff Development, IT Operations, Internet and e-Commerce system development and marketing program support, with staff of 10 IT professionals.

Implemented formal project management practises based on Prince2, to effectively plan, monitor, and assess the project status of large development projects.

Responded to the needs of the Executive business team with comprehensive planning, technology, and resources required to achieve their financial and operating goals.

Negotiated, purchased, and integrated a \$2 million e-Commerce platform. Fully operational

Identified numerous areas that lacked processes, and instituted procedures to ensure effective implementation, inclusive of; Streamlined processes for first-level operations and internal customer communication channels.

Nationwide WAN upgrade 80+ POP's full redundant MPLS core network.

GMAIL migration for College students scripting, strategy and implementation 15,000+ users.

Architected and implemented an outsourced managed Data Centre solution to absorb 160 servers into 10 physical machines.

Solution Architect

Nov 2006 – Jan 2007

Fulltime Employee.

Principal Architect and Network Engineer.

Proactively monitored server and network, ensuring 99.9% uptime. Provided 24/7 support during outages and critical production runs, Most hotels ran a 24/7 check in so uptime and support was a business imperative.

Implemented all network connectivity using Cisco Routers, BGP, Juniper SSL, MPLS, and Cisco PIX etc.

Architected redundant point of sale (POS) networks for both College and Hotel group training Fidelio systems.

Established and maintained disaster recovery procedures and documentation, including backup/recovery, offsite storage of backup tapes, fault-tolerant disk arrays with hot swappable drives and duplicate OS installations.

Conducted evaluations of existing technical documents for accuracy, completeness, and developed a penetration testing and vulnerability analysis plan.

Developed robust proxy and Internet Billing gateway for over 15,000 students and enterprise users, based on SQUID.

IMPEX PERSONNEL

Sydney, NSW, Australia

An International Transportation & Logistics Recruitment Agency, managing all facets of permanent, temporary and contract staffing across all aspects for Importers, Exporters, Supply Chain Logistics, Customs Brokerage, Freight Forwarding, Shipping Lines, Commodity Trading, Aviation, Transport

Warehousing, Sales & Marketing, Management and Administration operating in Australia and New Zealand and is headquartered in Sydney, New South Wales, Australia with annual revenues over AUD \$10 million, with staff of 300.

Head Network Engineer/ Senior Developer

Dec 2004 – Oct 2006

Fulltime Employee.

I.T Manager / Network Engineer and Member of Executive Leadership team.

Managed and performed statistical and tactical aspects on ImPex's WAN including but not limited to disaster recovery, auditing, security policies, procedures, guidelines and rollouts.

Implemented an Enterprise wide Terminal Services solution including a complex print solution, Staff would frequently travel between offices this solution would both route fax's automatically to the staff member's current location and allow print control per site.

I.T Business development and alignment with strategic goals.

Performed security administration functions (maintaining users, groups, ID's, and passwords) for HP, and UNIX systems at corporate and field locations.

Implemented VPN access with two-factor authentication, supporting remote users while meeting business security requirements. Deployed FTP services using SSL/ 128-bit encryption to handle sensitive candidates uploads/transfers.

TERTIARY EDUCATION

Bachelor of Business, Majoring in Information Technology (Business Information Systems)
Macquarie University, Sydney Australia (2008 Grad).

CERTIFICATION/ ADVANCED TRAINING

The Open Group Architecture Framework (TOGAF)

Microsoft Certified

- ◆ 6416B – Network Infrastructure & AD Technology skills Windows Server 2008
- ◆ 2274 – Managing Windows Server 2003 Environment
- ◆ MCSE NT4 and 2000 Server Full certified and recognized Systems Engineer
- ◆ MCSD + SQL
- ◆ 2433 – Microsoft Visual Basic Scripting Edition, Script Host Essentials
- ◆ 2576 – Implementing and Administering IIS 6.0
- ◆ MCSA 200+ Security, Comp TIA Security+ Operations Optimisation

Cisco Certified

- ◆ CCNE (Certified Network Engineer).
- ◆ CCIE (Held until 2009)
- ◆ CCNA (Certified Network Associate).

Citrix Certified

- ◆ Certified Administrator Citrix Access Gateway 4.

International Information Systems Security Certified

- ◆ ISC2 – Certified Information Systems Security Professional (CISSP)

Other Certification

- ◆ Ruxcon 2003 Computer Security Certification.
- ◆ Blackberry Enterprise Server Administrator course 4.0.
- ◆ IMR GOLD Alchemy Database / WWW Certified Database Administrator + Script.

TECHNICAL PROFILE - (SHORT LIST)

Methodologies –

PMBOK, Prince2, Agile (Scrum),

Frameworks – ITIL, TOGAF 9.0

PERSONAL

Residency Status – Australian Citizen.

PROFESSIONAL REFERENCES AVAILABLE ON REQUEST.